

```
humann3 -v -i ${root_dir}/analysis/humann3/${cur_sample_id}.fastq --threads 20 -o ${root_dir}/analysis/humann3/output --  
taxonomic-profile ${profile} --nucleotide-database ${database}/humann3_DB/chocophlan --protein-database  
${database}/humann3_DB/uniref
```

```
humann_renorm_table --input $file --output ${OUTPUT_DIR}/norm/${samplename}_cpm.tsv --units cpm --update-snames
```

```
humann_join_tables --input ${OUTPUT_DIR}/norm --output ${OUTPUT_DIR}/genefamilies_merge.tsv --file_name genefamilies_cpm
```

```
humann_join_tables --input ${OUTPUT_DIR}/norm --output ${OUTPUT_DIR}/pathabundance_merge.tsv --file_name pathabundance_cpm
```

```
humann_join_tables --input ${INPUT_DIR} --output ${OUTPUT_DIR}/pathcoverage_merge.tsv --file_name pathcoverage
```

```
humann_regroup_table --input ${OUTPUT_DIR}/genefamilies_merge.tsv --output ${OUTPUT_DIR}/genefamilies_merge_rxn.tsv  
--groups uniref90_rxn
```

```
humann_rename_table --input ${OUTPUT_DIR}/genefamilies_merge_rxn.tsv --output  
${OUTPUT_DIR}/genefamilies_merge_rxn_metacyc.tsv --names metacyc-rxn
```

```
humann_split_stratified_table --input ${OUTPUT_DIR}/genefamilies_merge_rxn_metacyc.tsv --output ${OUTPUT_DIR}/
```

```
humann_split_stratified_table --input ${OUTPUT_DIR}/pathabundance_merge.tsv --output ${OUTPUT_DIR}/
```

```
humann_split_stratified_table --input ${OUTPUT_DIR}/pathcoverage_merge.tsv --output ${OUTPUT_DIR}/
```

Humann3 Output

Normalisation



116_24mo_genefamilies.tsv
116_24mo_pathabundance.tsv
116_24mo_pathcoverage.tsv
...

Three tables per sample

116_24mo_genefamilies_cpm.tsv
116_24mo_pathabundance_cpm.tsv
116_24mo_pathcoverage_cpm.tsv
...

Joining

genefamilies_merge.tsv
pathabundance_merge.tsv
pathcoverage_merge.tsv

Three tables: each column is one sample in each

Regrouping



genefamilies_merge_rxn.tsv

Renaming

Named individual
reactions (enzymes)

genefamilies_merge_rxn_metacyc.tsv

Splitting



genefamilies_merge_rxn_metacyc_stratified.tsv
pathabundance_merge_stratified.tsv
pathcoverage_merge_stratified.tsv

genefamilies_merge_rxn_metacyc_unstratified.tsv
pathabundance_merge_unstratified.tsv
pathcoverage_merge_unstratified.tsv